

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 4: Revision

1 Match each term with the appropriate description using an arrow.

a balanced equation	A atoms, ions and molecules involved in a reaction
b chemical bond	B both sides have equal numbers of each type of atom
c chemical equation	C description of a reaction using chemical symbols
d chemical species	D description of a reaction using words
e combination reaction	E electrostatic force of attraction holding atoms together
f combustion	F energy cannot be created or destroyed, only transformed
g conservation of energy	G equation that excludes any spectator ions
h conservation of mass	H equation that includes both chemicals and the energy change
i decomposition reaction	I equation that specifies all ions present in the reaction vessel
j endothermic reaction	J heat absorbed in a reaction at a constant pressure
k enthalpy	K heat needed to raise the temperature of 1 gram by 1 degree K
l exothermic reaction	L heat released when a fuel undergoes complete combustion
m heat of combustion	M ions that do not participate in a reaction
n ionic equation	N ions that take an active part in a reaction
o net ionic equation	O new substances formed in a chemical reaction
p participant ions	P when the number of each type of atom is the same before as after
q products	Q occurs when a compound breaks down into simpler chemicals
r reactants	R occurs when atoms of two or more elements combine
s reaction	S occurs when old substances are transformed into new ones
t specific heat capacity	T old substances consumed in a chemical reaction
u spectator ions	U reaction in which oxygen reacts with a fuel to give an oxide
v spontaneous reaction	V reaction that occurs at room temperature without any heating
w symbol equation	W reaction that releases heat to the environment
x thermochemical equation	X reaction that requires energy to sustain it

2 Explain the differences between the terms in each of the items below.

a Reactants and products

b Chemical and symbol equations

c Combination and decomposition reactions

d Exothermic and endothermic reactions

e Ionic and thermochemical equations

f Heat of combustion and specific heat capacity

3 Decide whether each of the following statements is true or false. If the statement is false, correct it.

a The arrow in a chemical equation always points in the direction of the reaction. (True / False)

b In a chemical reaction, reactants are transformed into products. (True / False)

c For an equation to be balanced, the number of atoms must be the same on both sides of the equation. (True / False)

- d** A balanced equation ensures that the law of conservation of energy holds. (True / False)

- e** In a symbol equation, if a species does not have a specified state then it is assumed to be in an aqueous state. (True / False)

- f** Chemicals react in order to achieve chemical stability. (True / False)

- g** Energy is released when bonds are broken and is absorbed when bonds are formed. (True / False)

- h** Spontaneous reactions do not require any energy input to start. (True / False)

- i** Ionic equations always include only those ions that participate in the reaction. (True / False)

- j** If less energy is stored in the bonds of the products than the reactants, an exothermic reaction must have occurred. (True / False)

- k** The heat of reaction of an endothermic reaction is always negative. (True / False)

- l** The temperature of 1 kg of different substances will always increase by the same amount if the same amount of heat energy is supplied. (True / False)

- 4 Complete the following table to explain why energy is required to break the bonds found in the following substances.

Type of bond	Nature of bonding	Reason why energy is required
Metallic bonding	Attraction between delocalised electrons and metal cations	
Ionic bonding	Attraction between anions and cations in a lattice structure	
Covalent bonding	Attraction between nuclei of neighbouring non-metal atoms and shared electrons	

- 5 Study the thermochemical equation for the reaction between magnesium and oxygen gas.

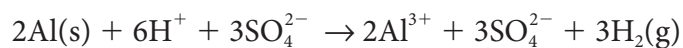


a Is it a balanced equation or not? How do you know?

b Is it an exothermic or endothermic reaction? How do you know?

c Is it a spontaneous reaction or not? How do you know?

- 6 Study the ionic equation for the reaction between aluminium and sulphuric acid.



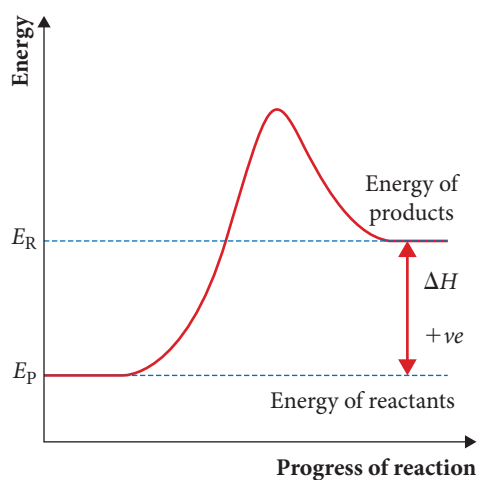
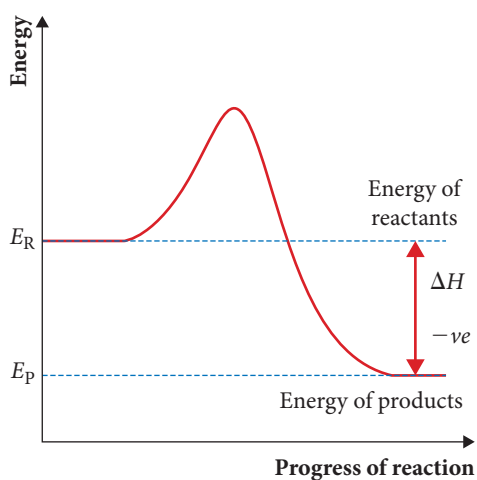
a Is it a balanced equation or not? How do you know that?

b What state are the aluminium ions in? How do you know that?

c Which are the spectator ions? How do you know that?

d Write the net ionic equation for the reaction.

7 The graphs below show the energy changes that occur during two different reactions. E_R and E_P stand for the enthalpy of the reactants and products respectively.



a What does a negative ΔH value mean?

b What does a positive ΔH value mean?

c Which reaction is the exothermic reaction? How do you know?

d Which reaction is the endothermic reaction? How do you know?

e Will the exothermic reaction occur spontaneously or not? How do you know?

8 The formula for calculating the amount of heat, q , involved when a substance is heated is:

$$q = mC\Delta T$$

a What do each of the quantities on the right side of the equation stand for?

b What units must the quantities on the right side be in?

c What units will heat (q) be in?

d Why can ΔT be measured in degrees Celsius ($^{\circ}\text{C}$)?